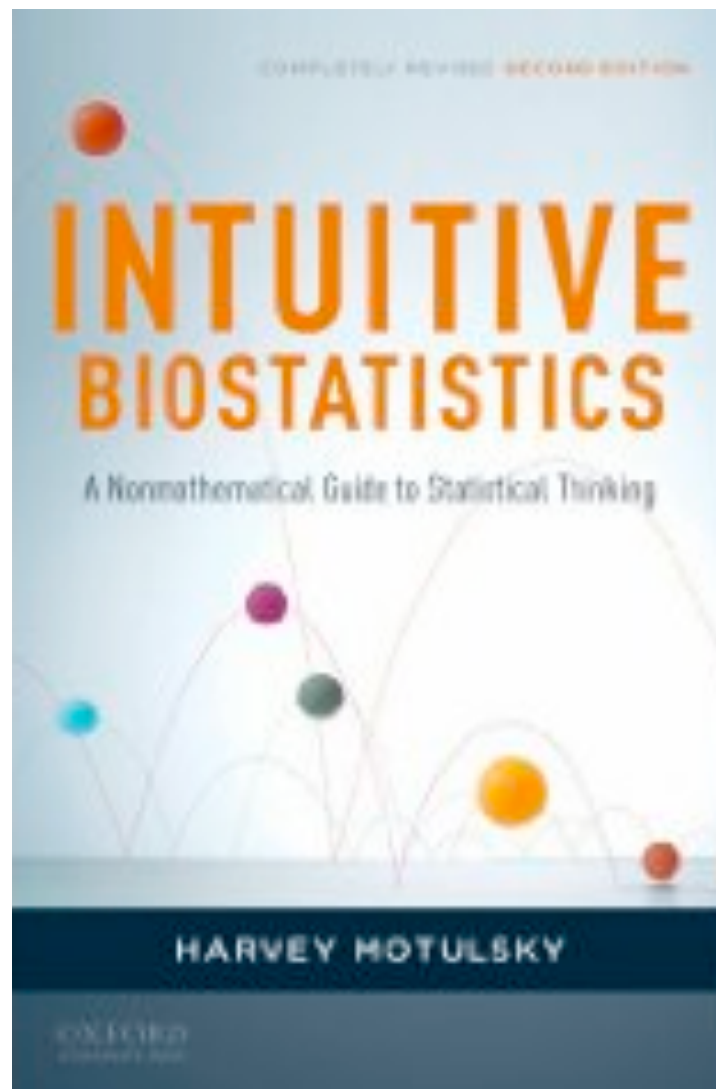


Statistik in der Medizin

Von Daten zu Entscheidungen

Tobias Straub, BMC, LMU

**Warum
sind wir freiwillig hier?**



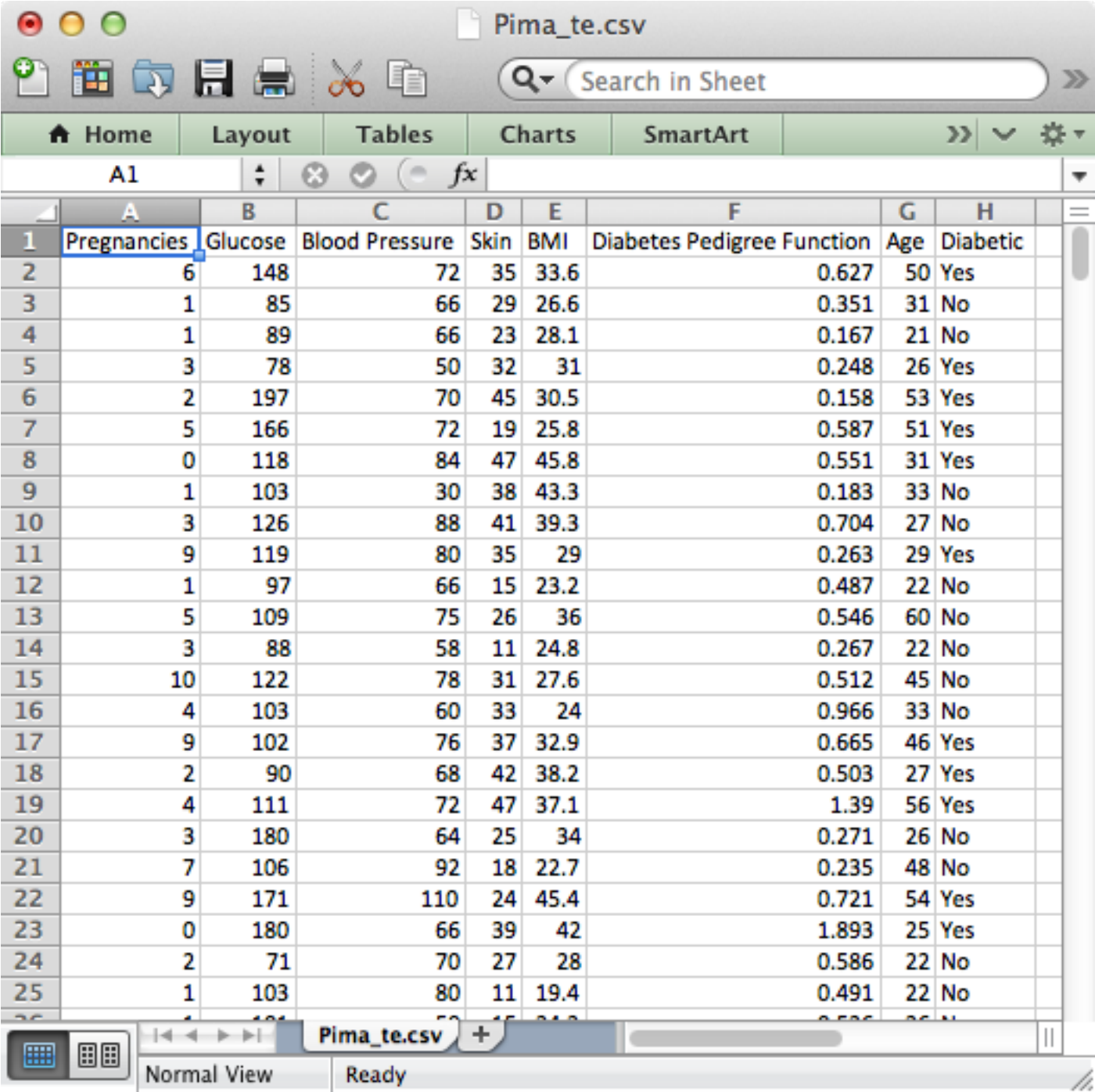
chatGPT

Worum geht's?

- Messdaten
- Effekte
- Unsicherheit
- Entscheidung

Daten und Metadaten

Daten

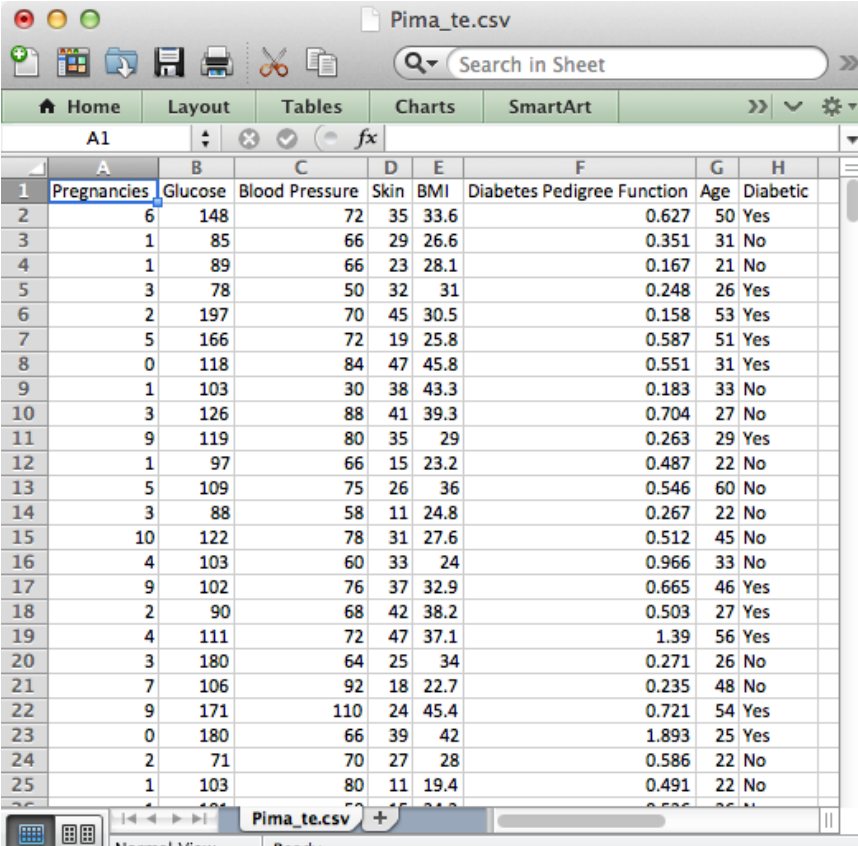


The screenshot shows a spreadsheet application window titled "Pima_te.csv". The interface includes a menu bar with "Home", "Layout", "Tables", "Charts", and "SmartArt". A search bar labeled "Search in Sheet" is present. The spreadsheet itself has columns labeled A through H. The data is as follows:

	A	B	C	D	E	F	G	H
1	Pregnancies	Glucose	Blood Pressure	Skin	BMI	Diabetes Pedigree Function	Age	Diabetic
2	6	148	72	35	33.6	0.627	50	Yes
3	1	85	66	29	26.6	0.351	31	No
4	1	89	66	23	28.1	0.167	21	No
5	3	78	50	32	31	0.248	26	Yes
6	2	197	70	45	30.5	0.158	53	Yes
7	5	166	72	19	25.8	0.587	51	Yes
8	0	118	84	47	45.8	0.551	31	Yes
9	1	103	30	38	43.3	0.183	33	No
10	3	126	88	41	39.3	0.704	27	No
11	9	119	80	35	29	0.263	29	Yes
12	1	97	66	15	23.2	0.487	22	No
13	5	109	75	26	36	0.546	60	No
14	3	88	58	11	24.8	0.267	22	No
15	10	122	78	31	27.6	0.512	45	No
16	4	103	60	33	24	0.966	33	No
17	9	102	76	37	32.9	0.665	46	Yes
18	2	90	68	42	38.2	0.503	27	Yes
19	4	111	72	47	37.1	1.39	56	Yes
20	3	180	64	25	34	0.271	26	No
21	7	106	92	18	22.7	0.235	48	No
22	9	171	110	24	45.4	0.721	54	Yes
23	0	180	66	39	42	1.893	25	Yes
24	2	71	70	27	28	0.586	22	No
25	1	103	80	11	19.4	0.491	22	No

Ebenso wichtig: Metadaten

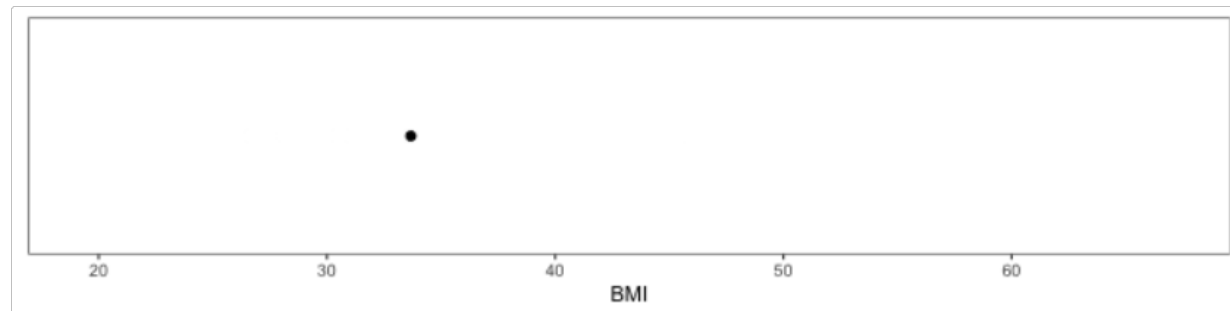
- Wie wurden die Daten gesammelt?
zB Beobachtungsstudie
- Unter welchen Bedingungen
wurden die Daten erhoben? Ort,
Datum, Person



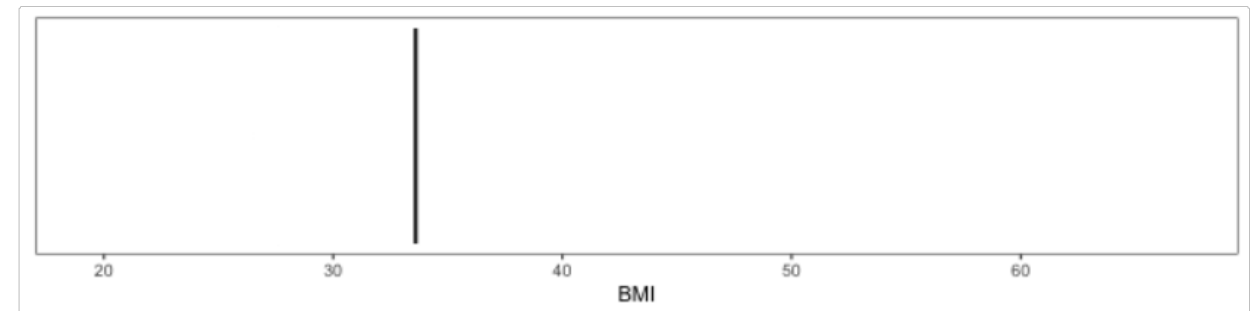
	A	B	C	D	E	F	G	H
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24	2	71	70	27	28	0.586	22	No
25	1	103	80	11	19.4	0.491	22	No

Visualisierung

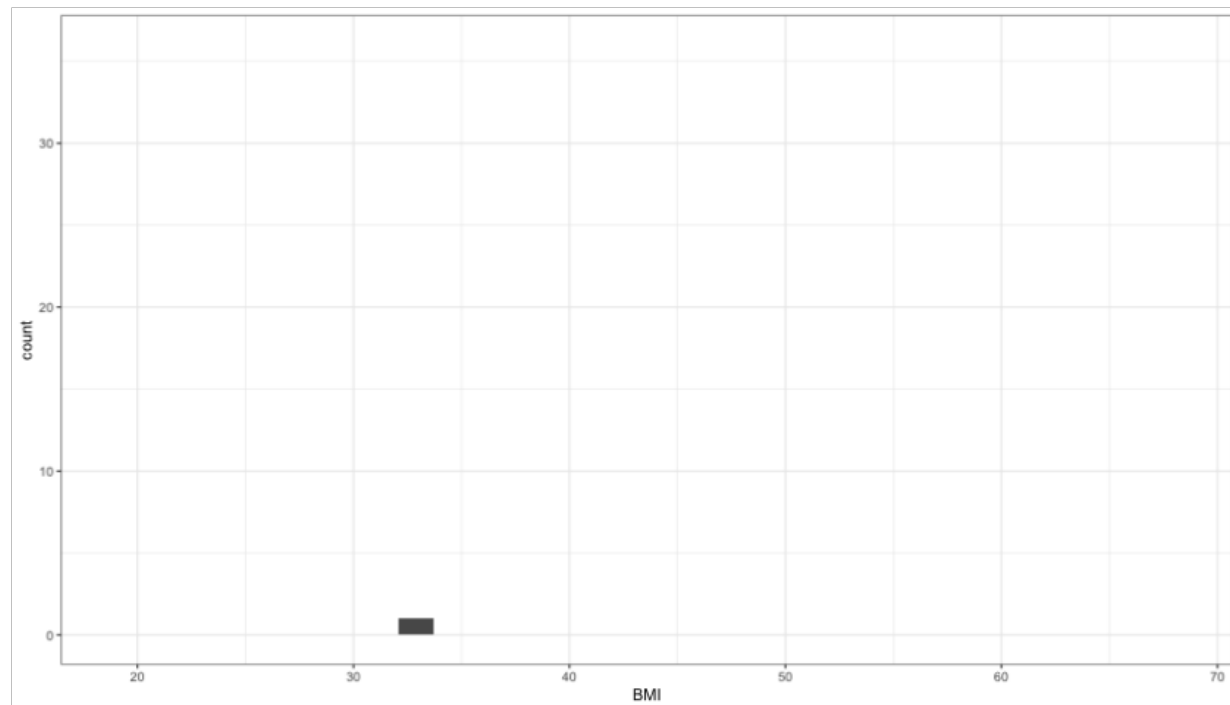
strip chart



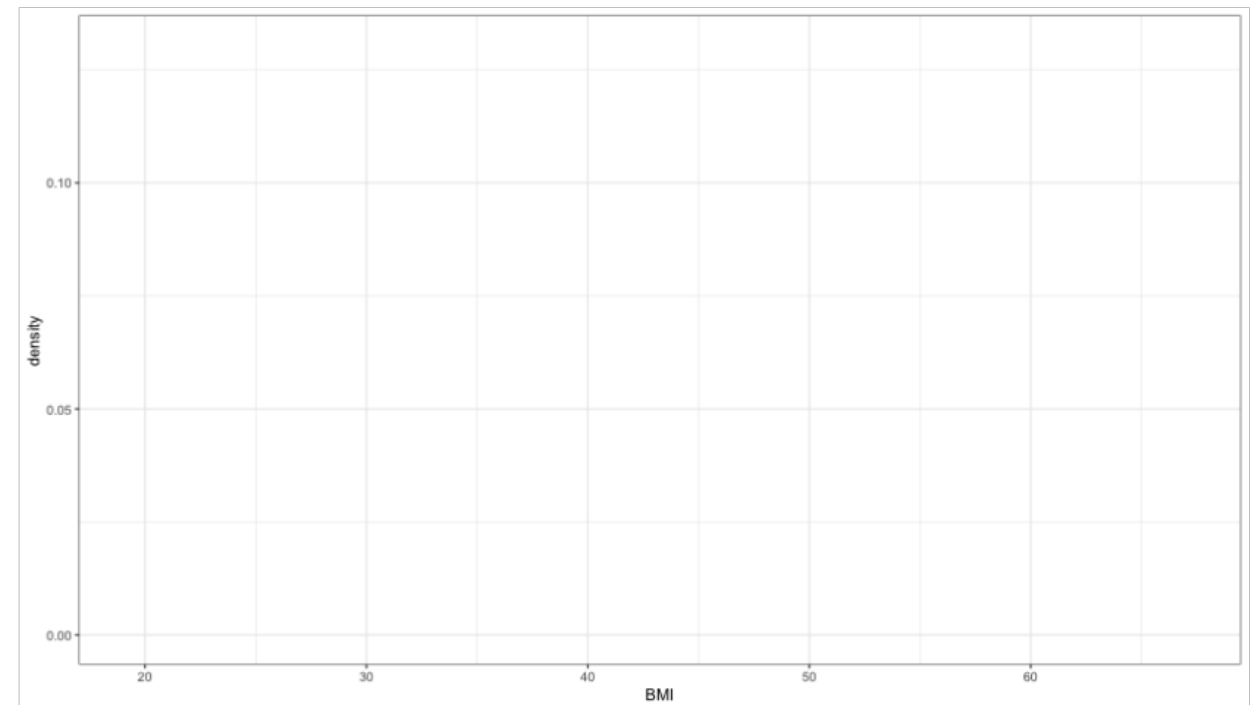
boxplot



histogram

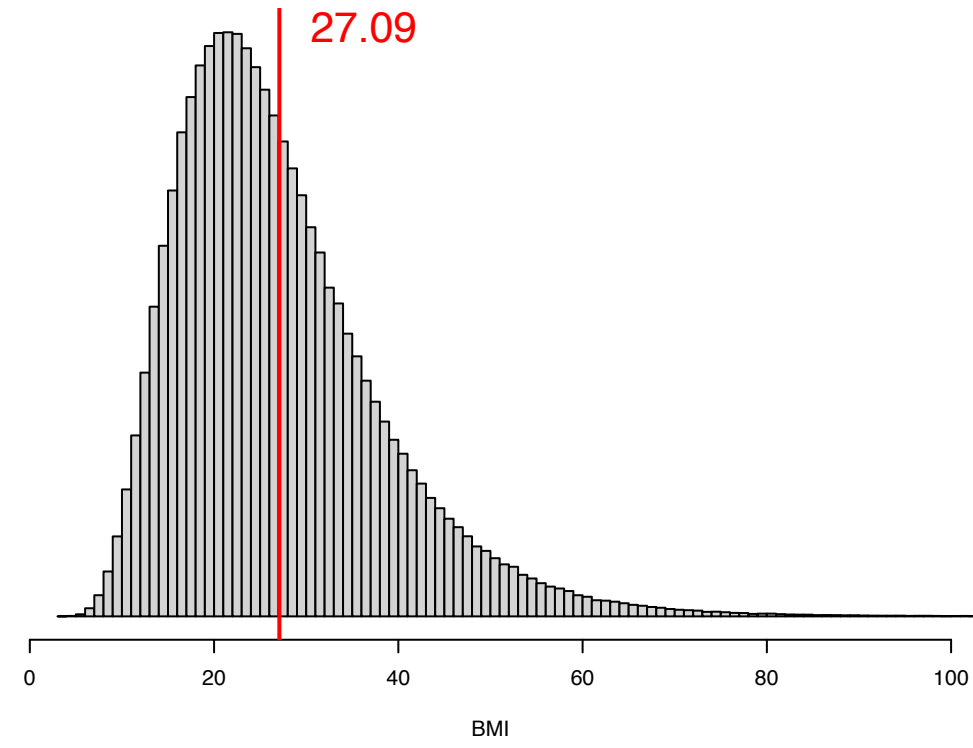


density plot



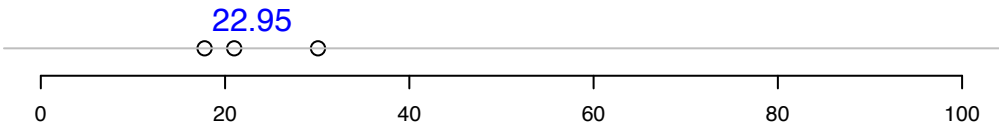
Population & Stichprobe

Sind meine Beobachtungen allgemein gültig?



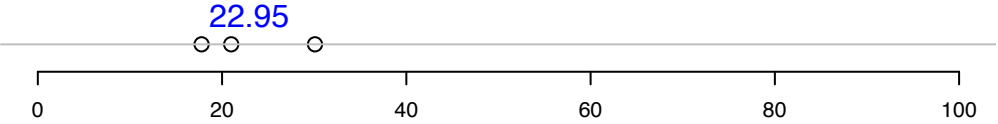
BMI

Study: US

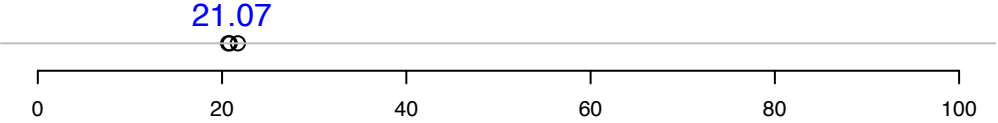


BMI

Study: US

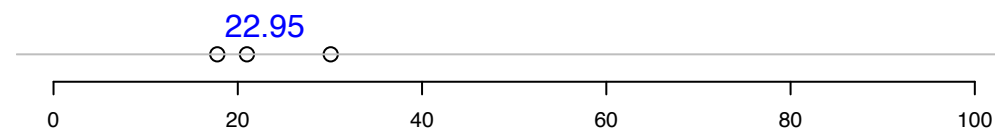


Study: Europe

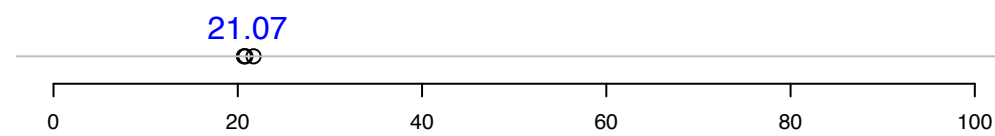


BMI

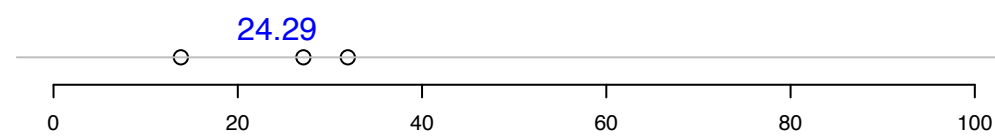
Study: US



Study: Europe



Study: Australia

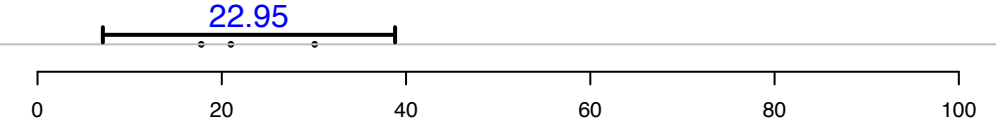


Konfidenzintervall

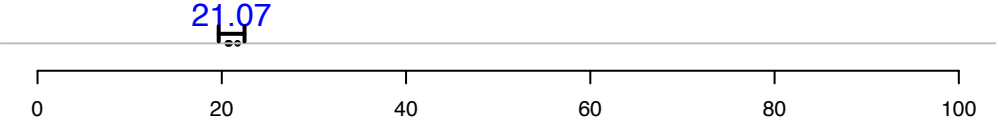
- ein Bereich von Werten, welcher mit einer bestimmten Wahrscheinlichkeit (zB 95%) den wahren Wert einer unbekannten statistischen Größe (wie dem Mittelwert) der Population einschließt. Wird aus der Stichprobe berechnet.
- Die Stichprobe sollte zufällig und repräsentativ für die Grundgesamtheit sein.

BMI

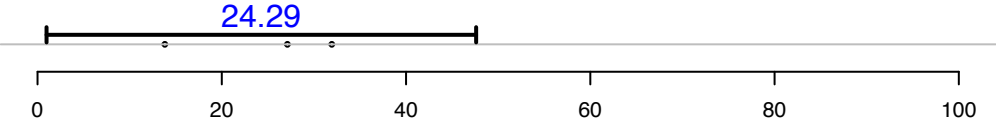
Study: US

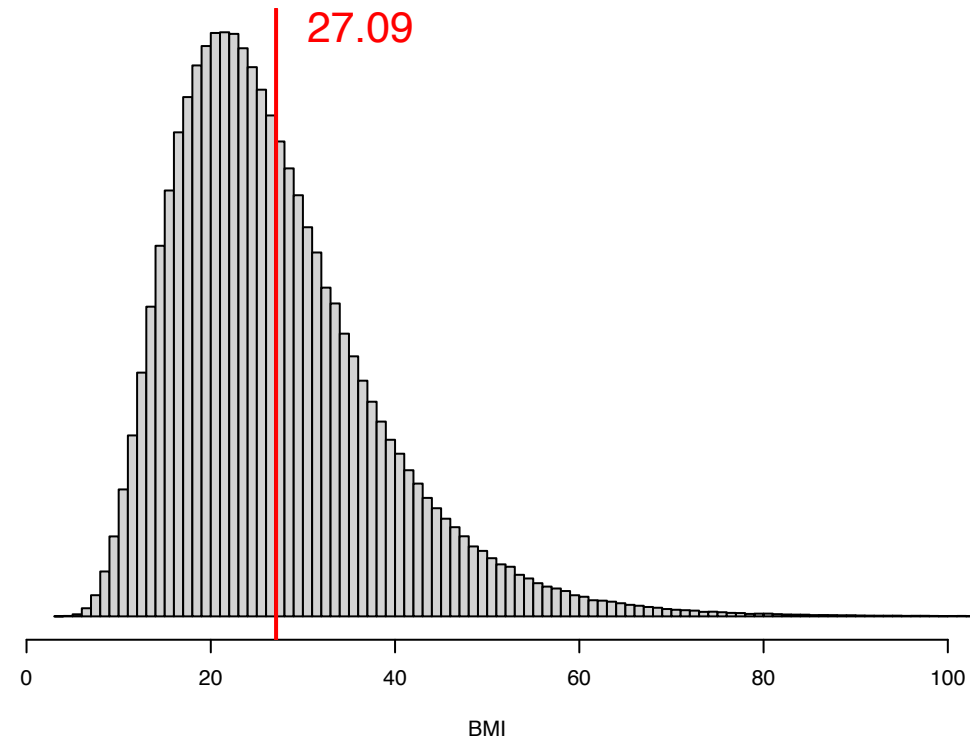


Study: Europe

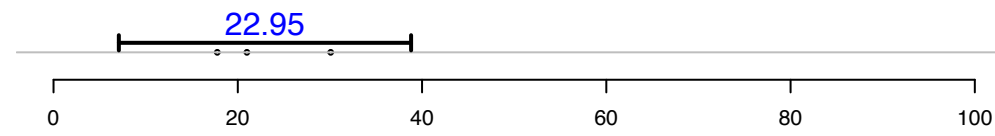


Study: Australia

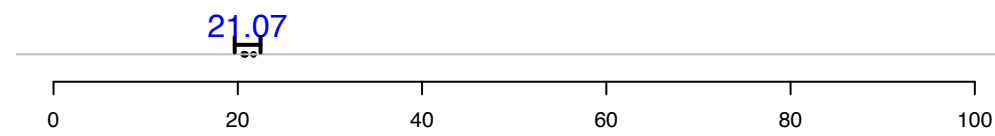




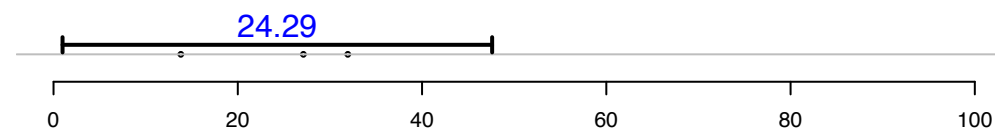
Study: US

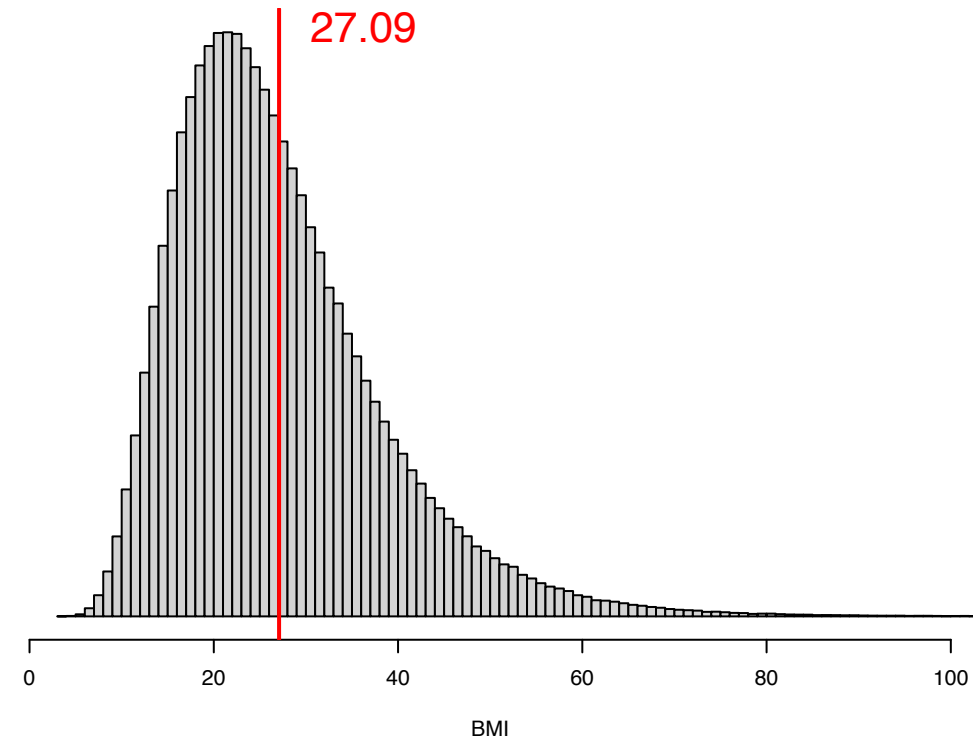


Study: Europe

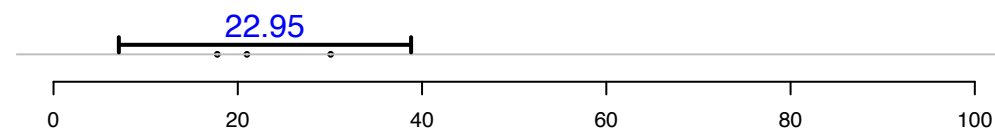


Study: Australia

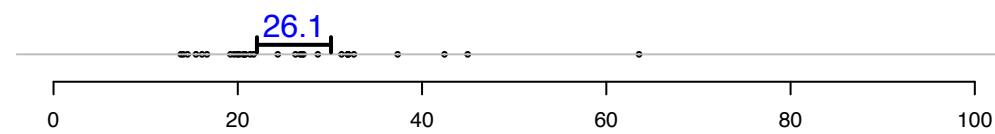




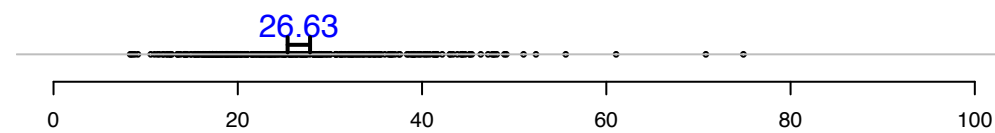
Study: Asia n=3



Study: Africa n=30



Study: Antarctica n=300



**Voraussetzung für Inferenz:
Repräsentativität**

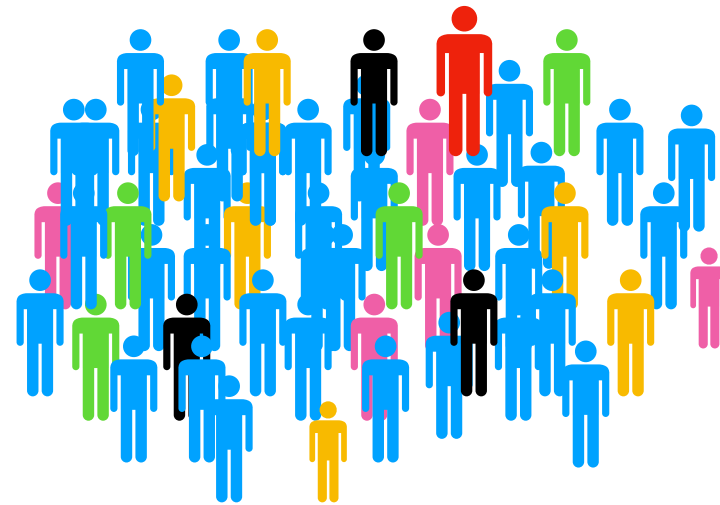
Kontrollierte Studie Experiment



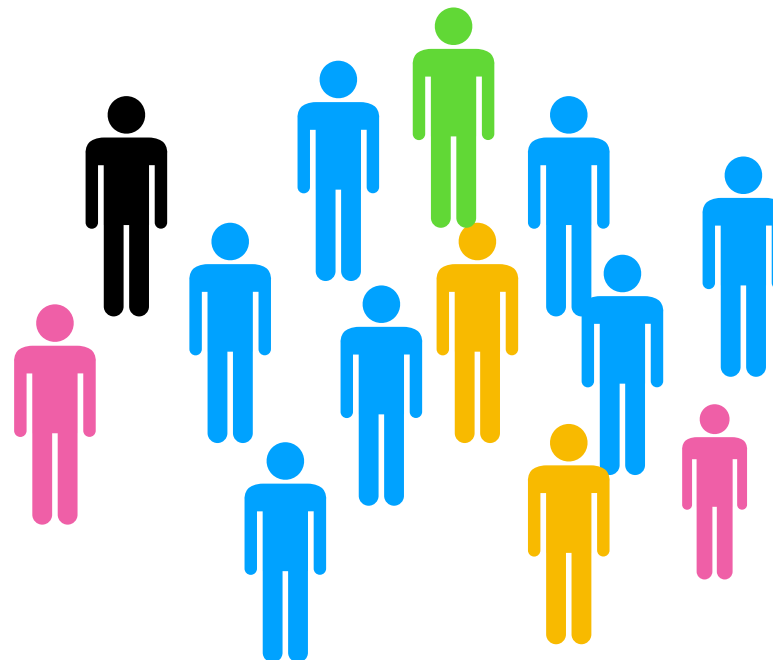
Wirkt die Diät?

Stichprobe

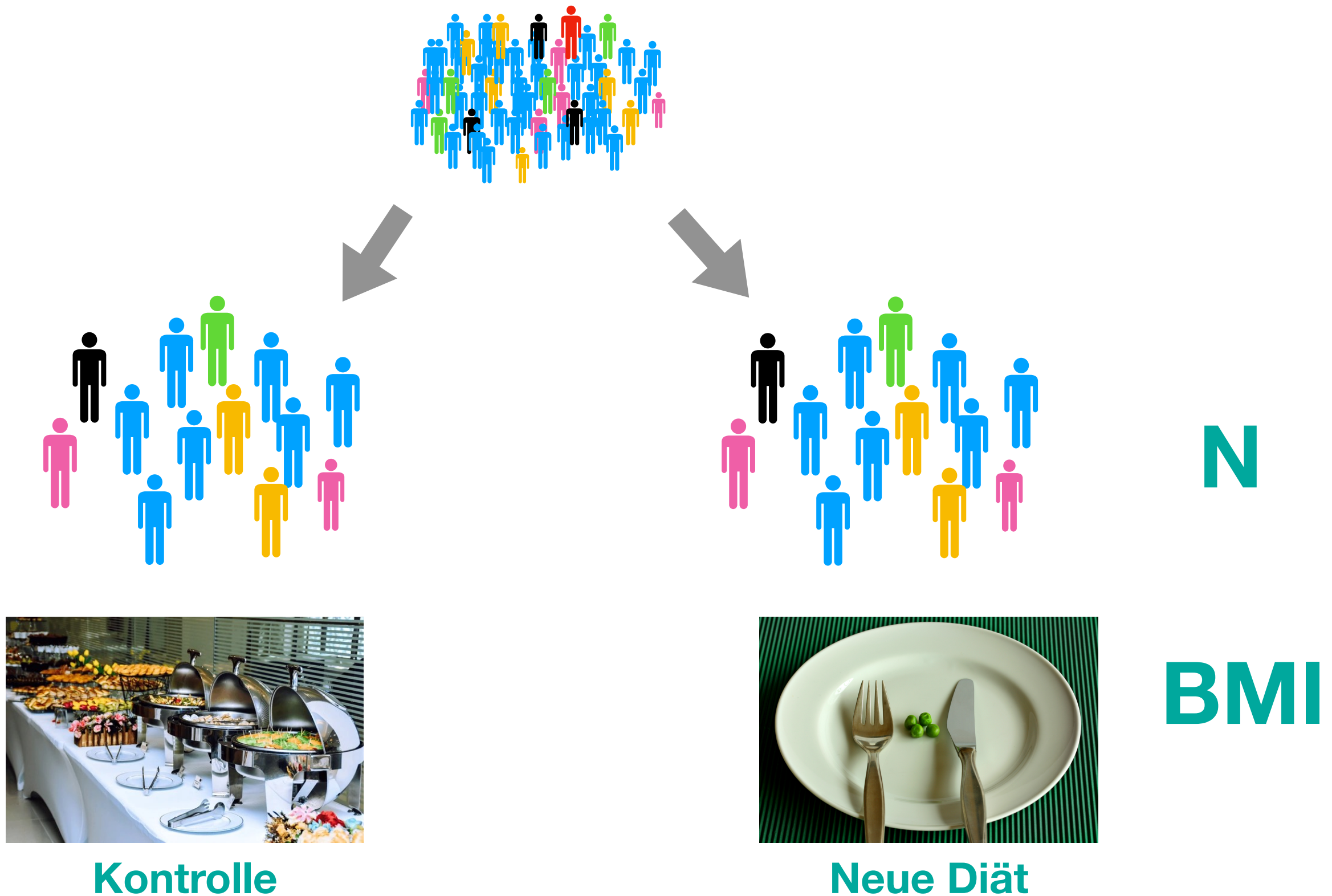
Gesamtpopulation
(Alle? Nur BMI > 30 ? Alter? ...)



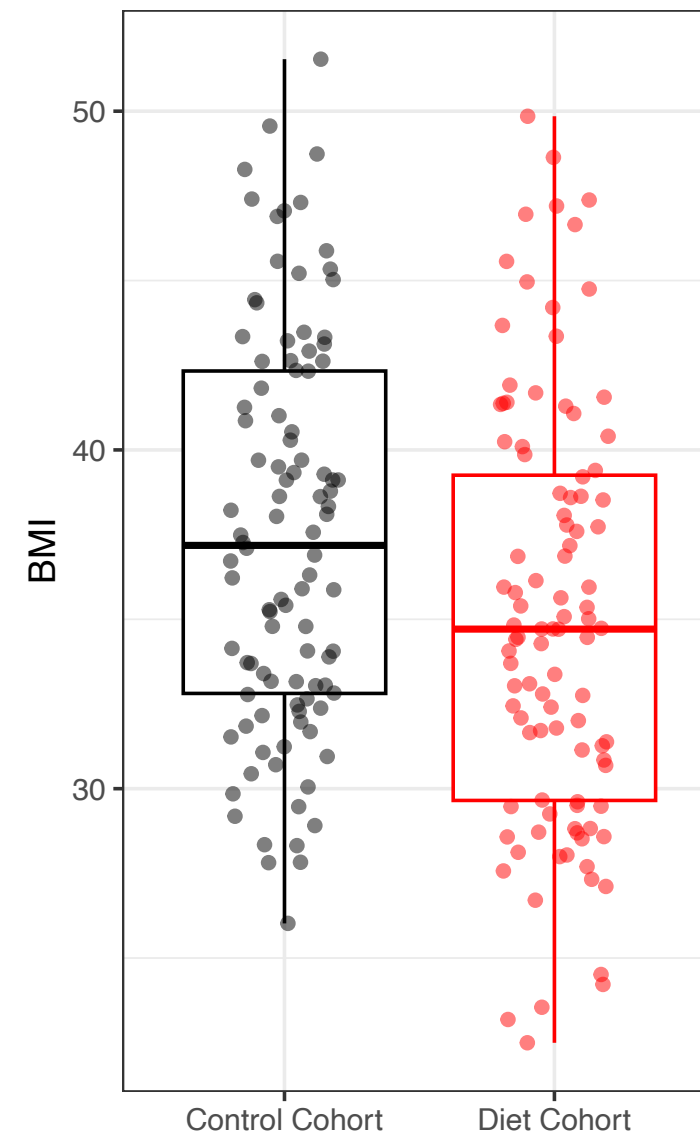
Repräsentative Stichprobe



Eine Studie



Ergebnis

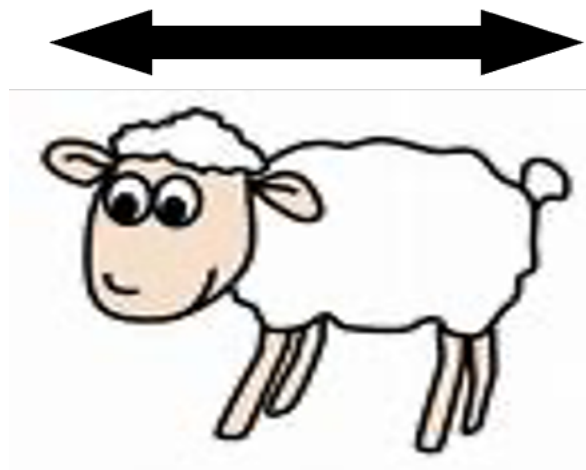


Robust?

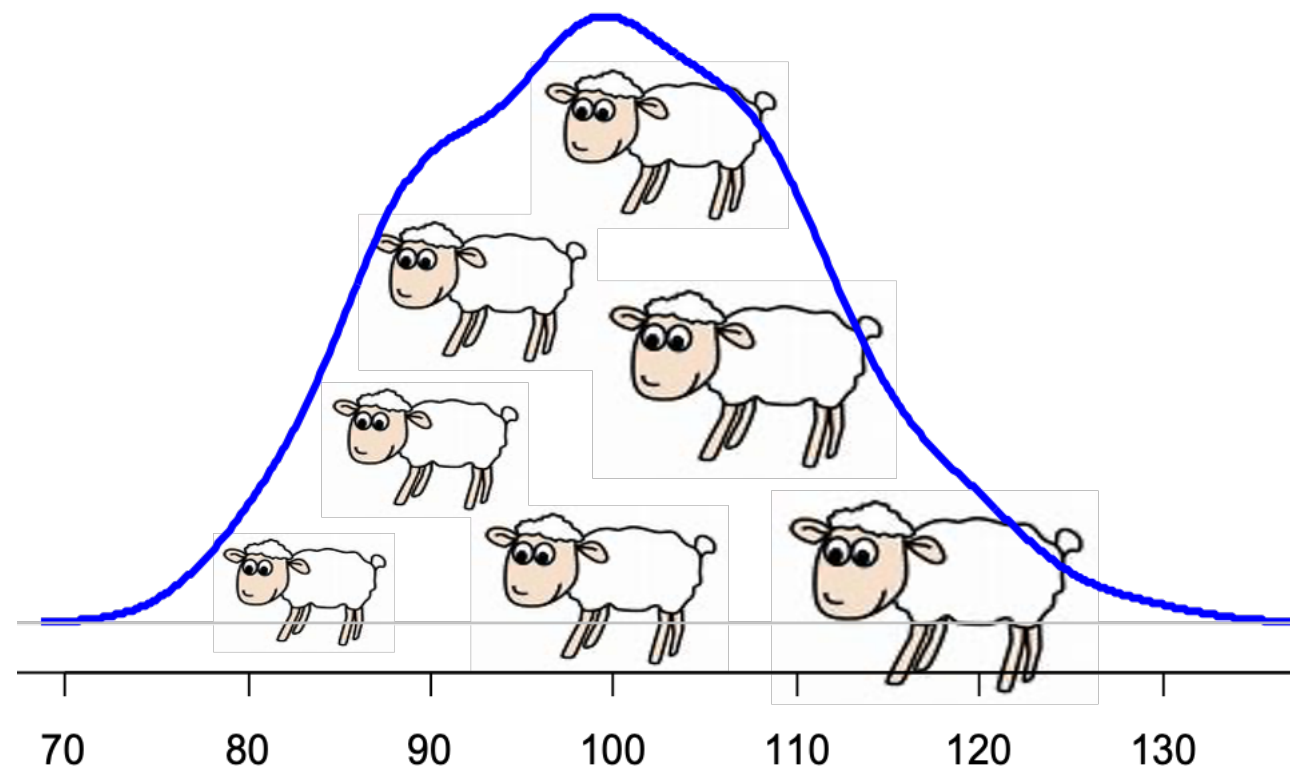
Wahrheit gibt es eine Wege dorthin viele @lakens

- **Frequentist** (Neyman Pearson):
 - Auswertung laufender Datenerhebung. Sagt nichts Konkretes über den aktuellen Test aus
- **Likelihood**
 - Vergleicht die Wahrscheinlichkeit verschiedener Hypothesen anhand der Daten.
- **Bayesian**
 - Aktualisierung bestehender Überzeugungen angesichts neuer Daten.

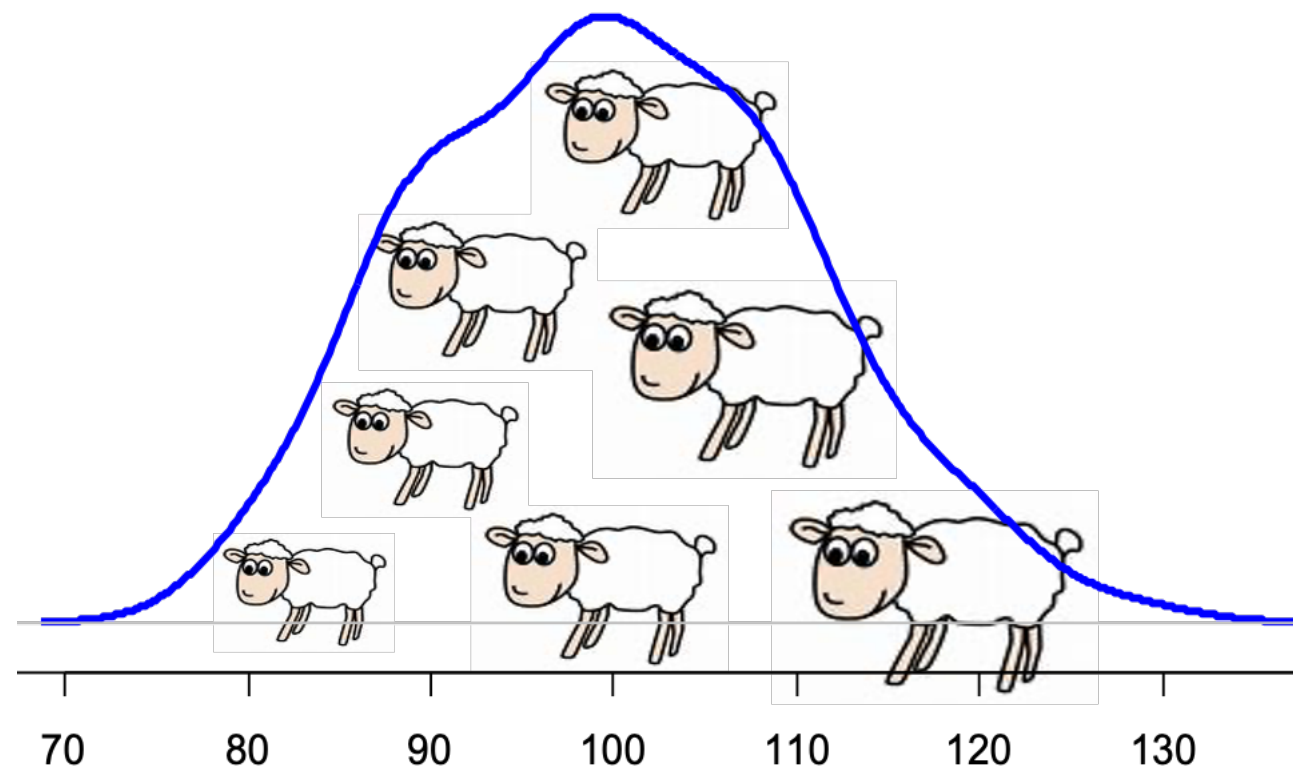
Ein non-sheep Detektor



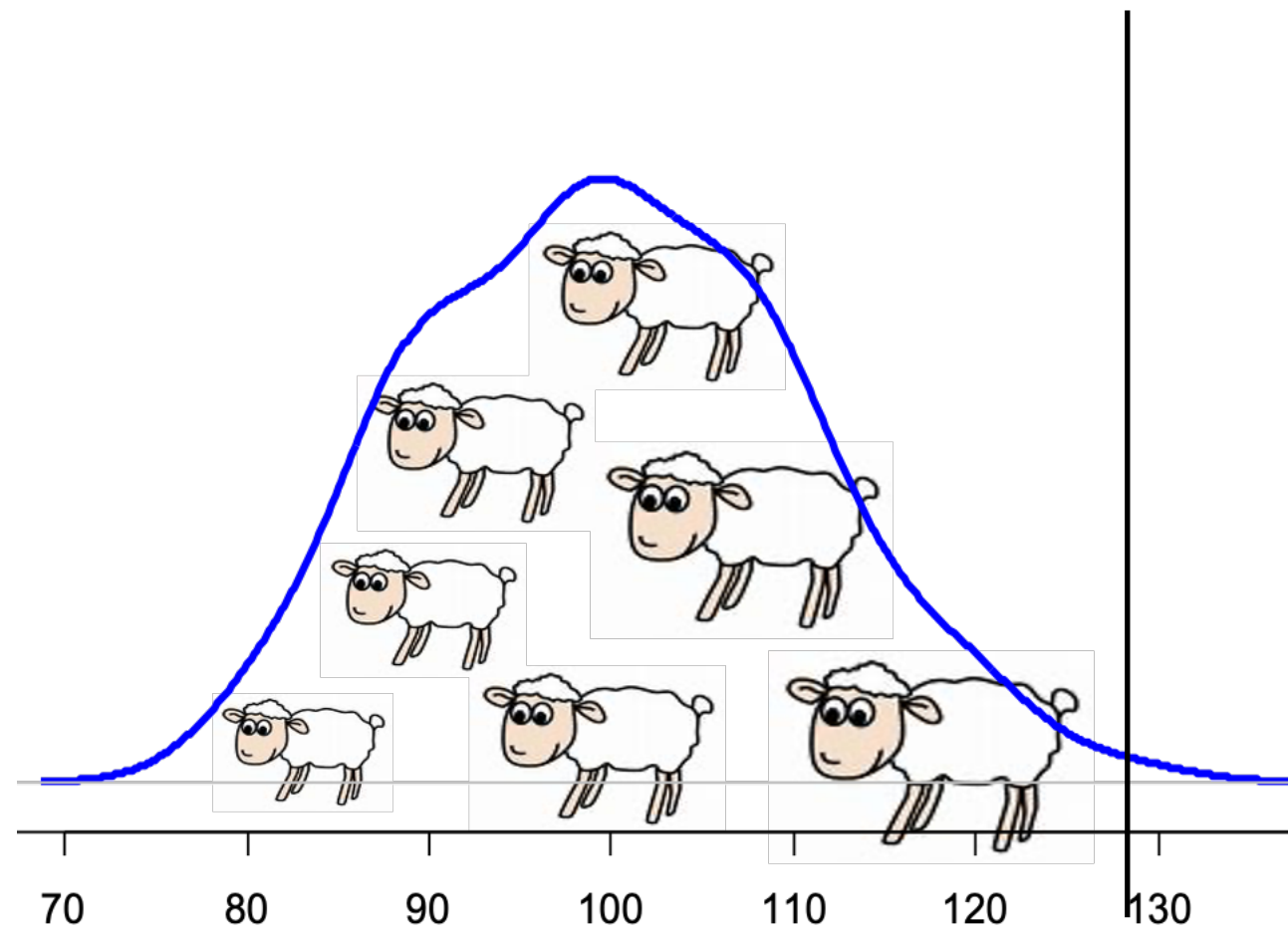
Ein non-sheep Detektor



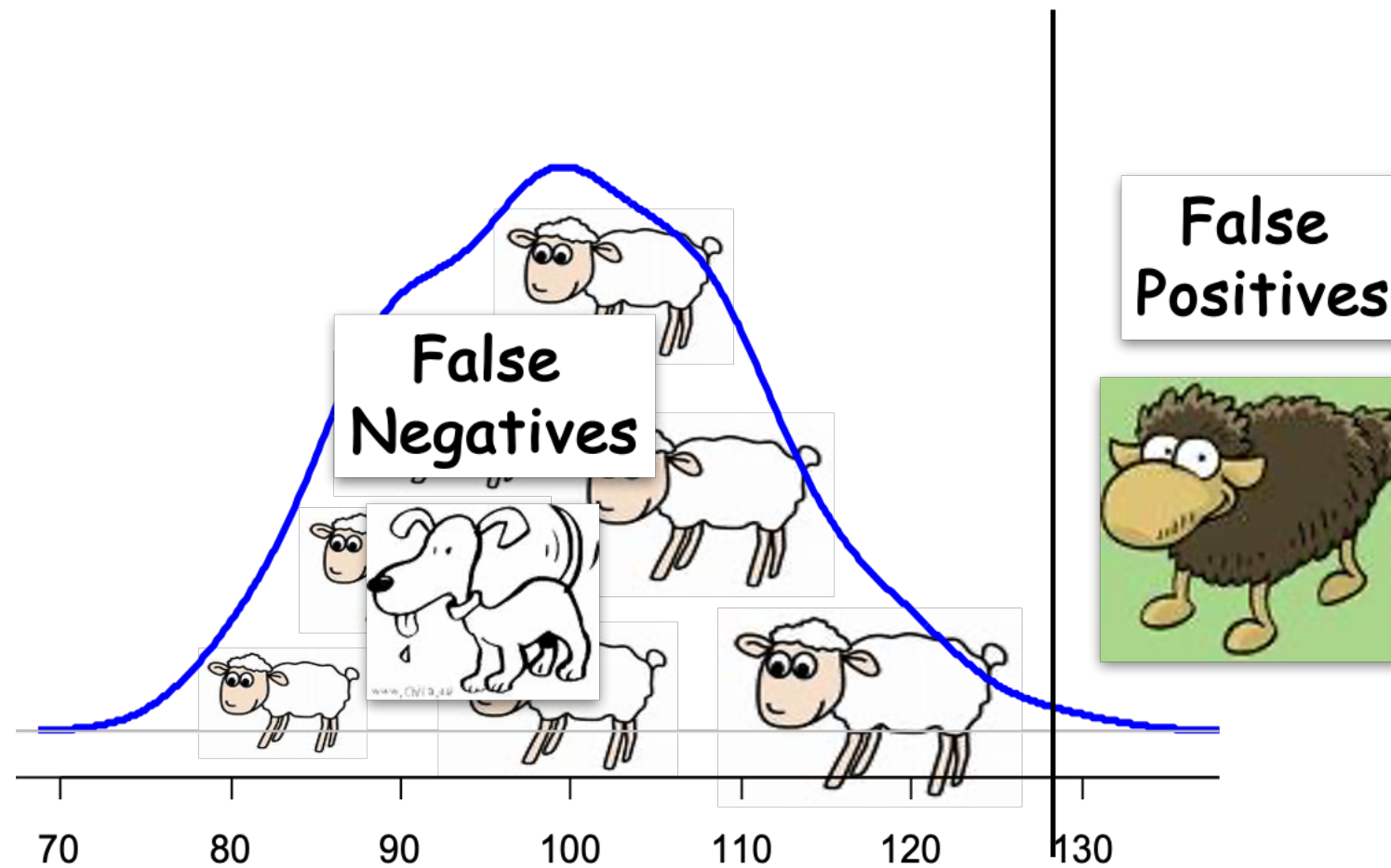
Ein non-sheep Detektor



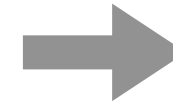
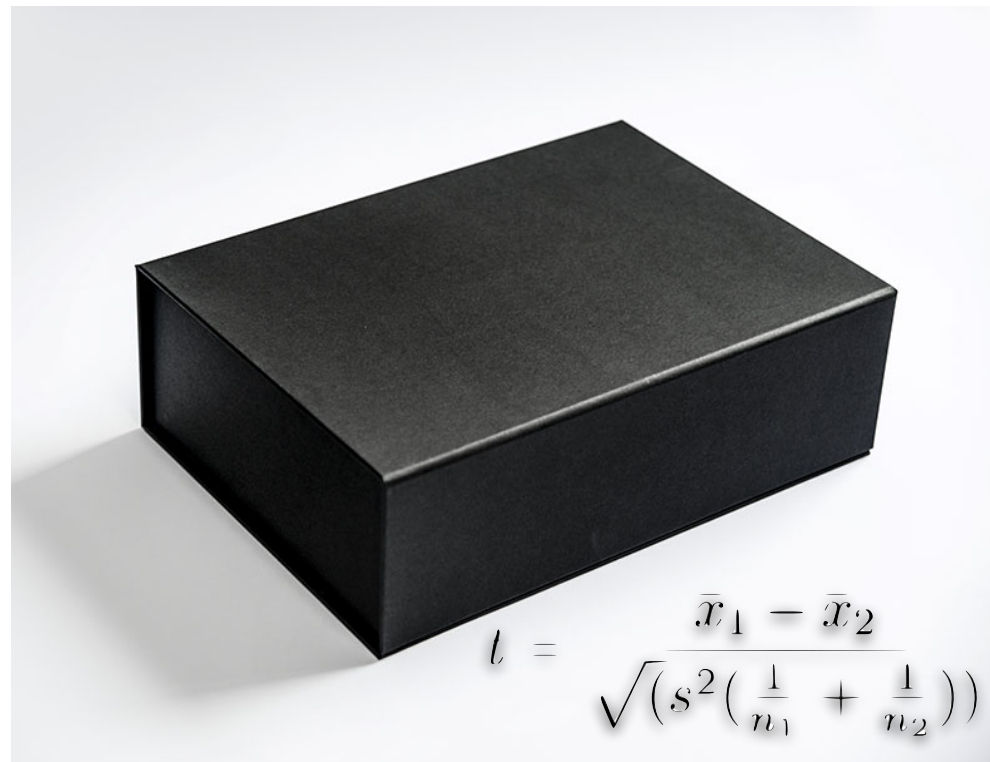
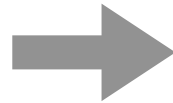
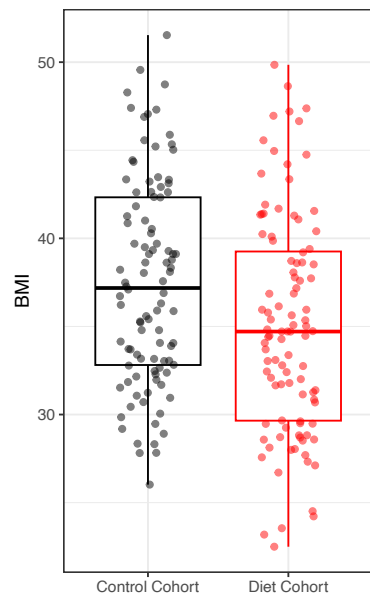
Ein non-sheep Detektor



Ein non-sheep Detektor



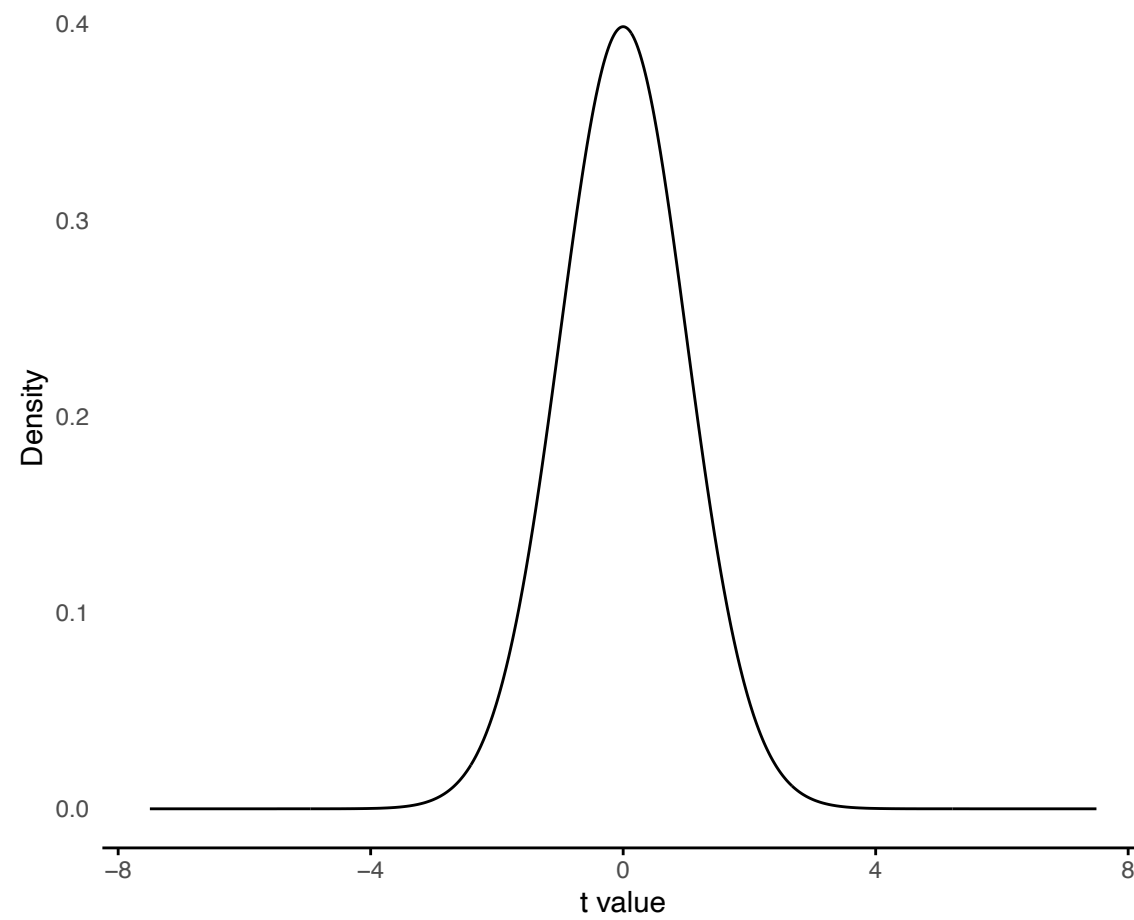
Statistisches Testen

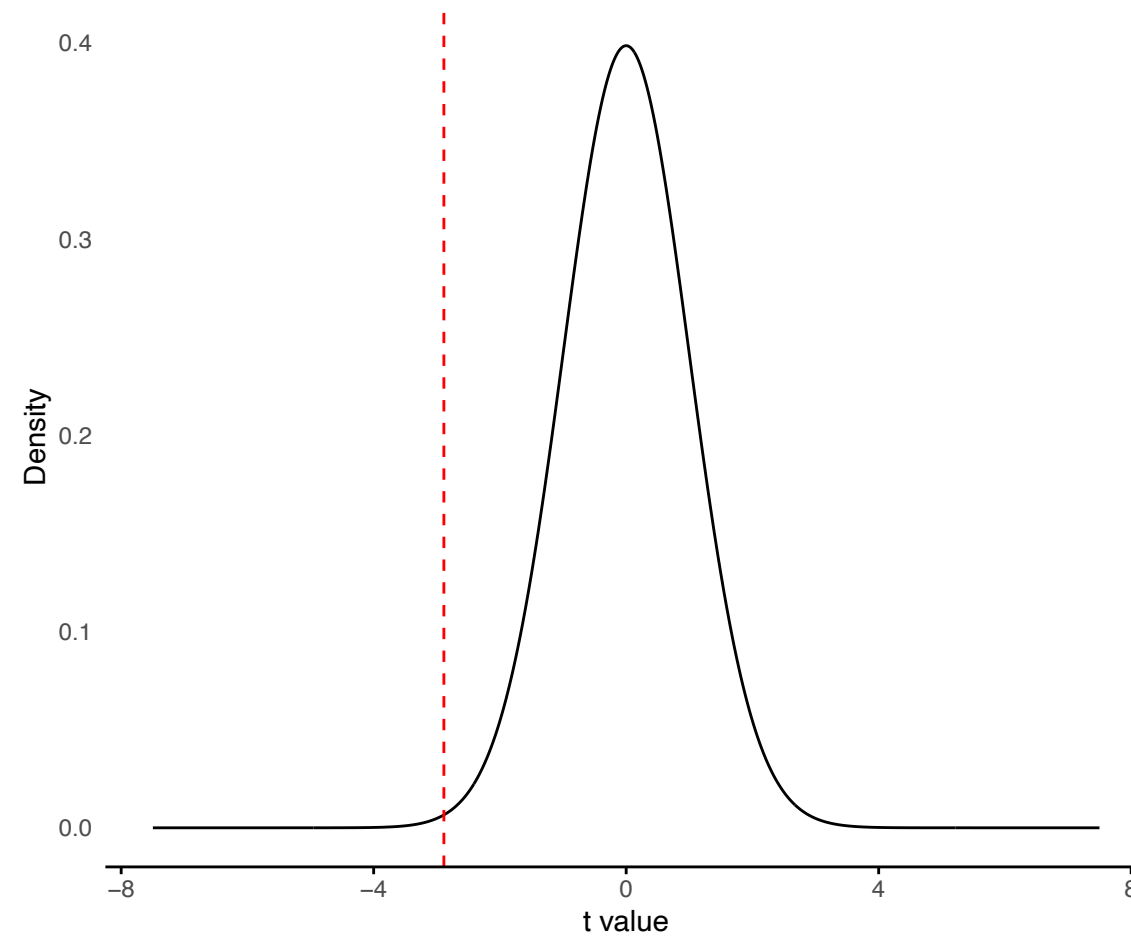


t

Statistisches Testverfahren

t-Test

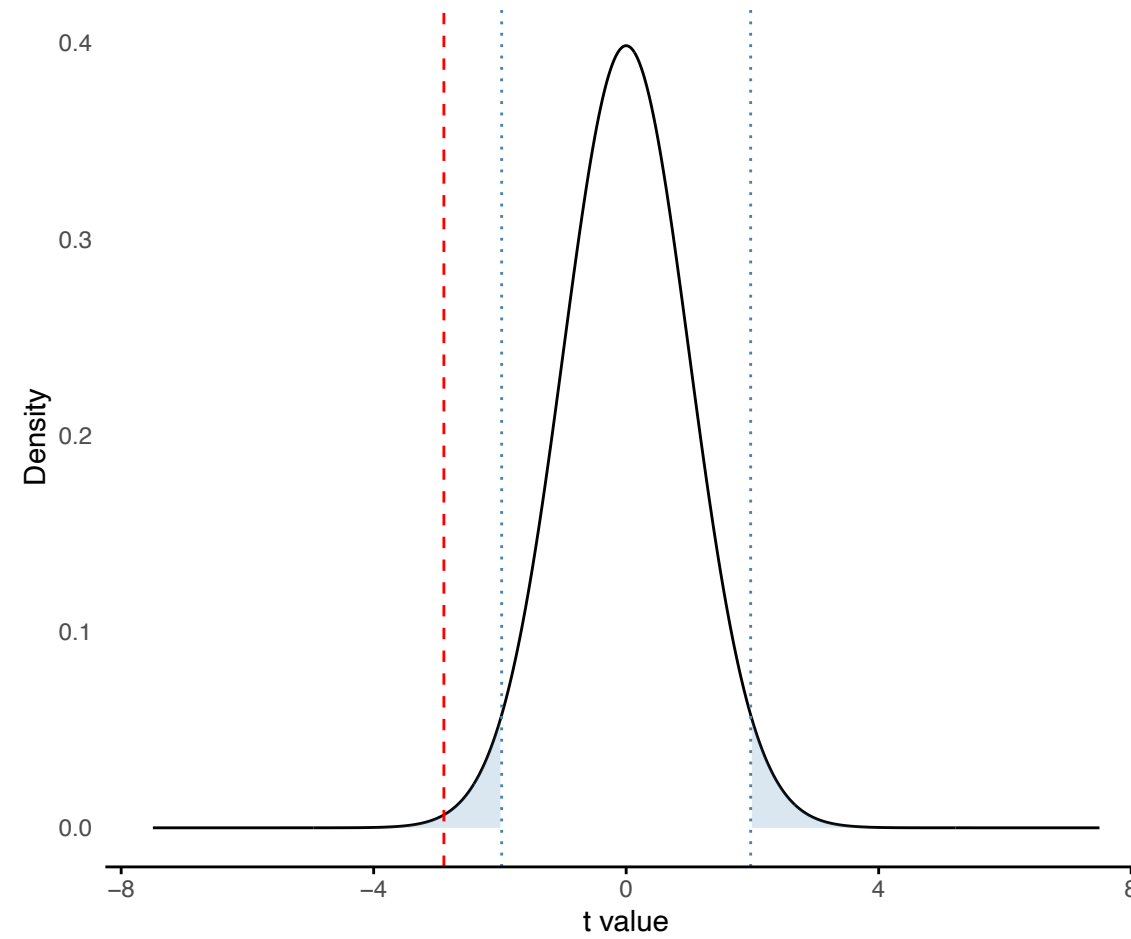




$t = -2.9$

Der p-Wert

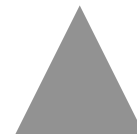
- Wahrscheinlichkeit (engl. *probability*, p) dass solche Unterschiede in der Stichprobe zufälliger Natur sind (=die Diät wirkt nicht=kein Effekt in der Gesamtpopulation)?
- kann aus t berechnet werden.
- Konvention: ist die Wahrscheinlichkeit kleiner als 5%,
sprechend wir von einem “signifikanten” Effekt.
- Kein zufälliger Effekt, “Die Diät wirkt!”
- Wichtig: ist die Wahrscheinlichkeit größer als 5% kann **nicht** behauptet werden, dass die Diät nicht wirkt.



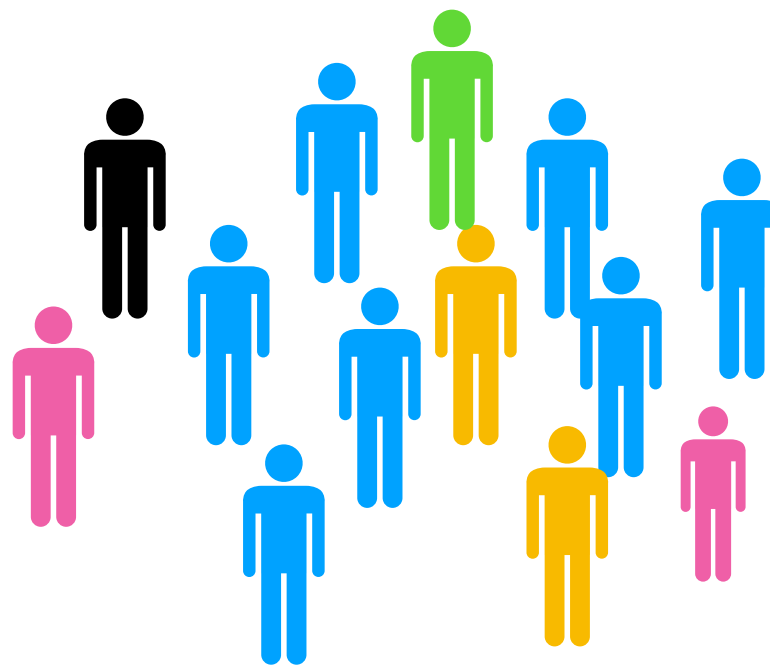
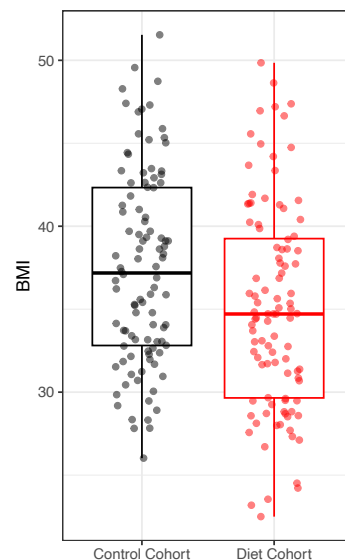
$t = -2.9$
p-value= 0.004

Konfidenzintervall

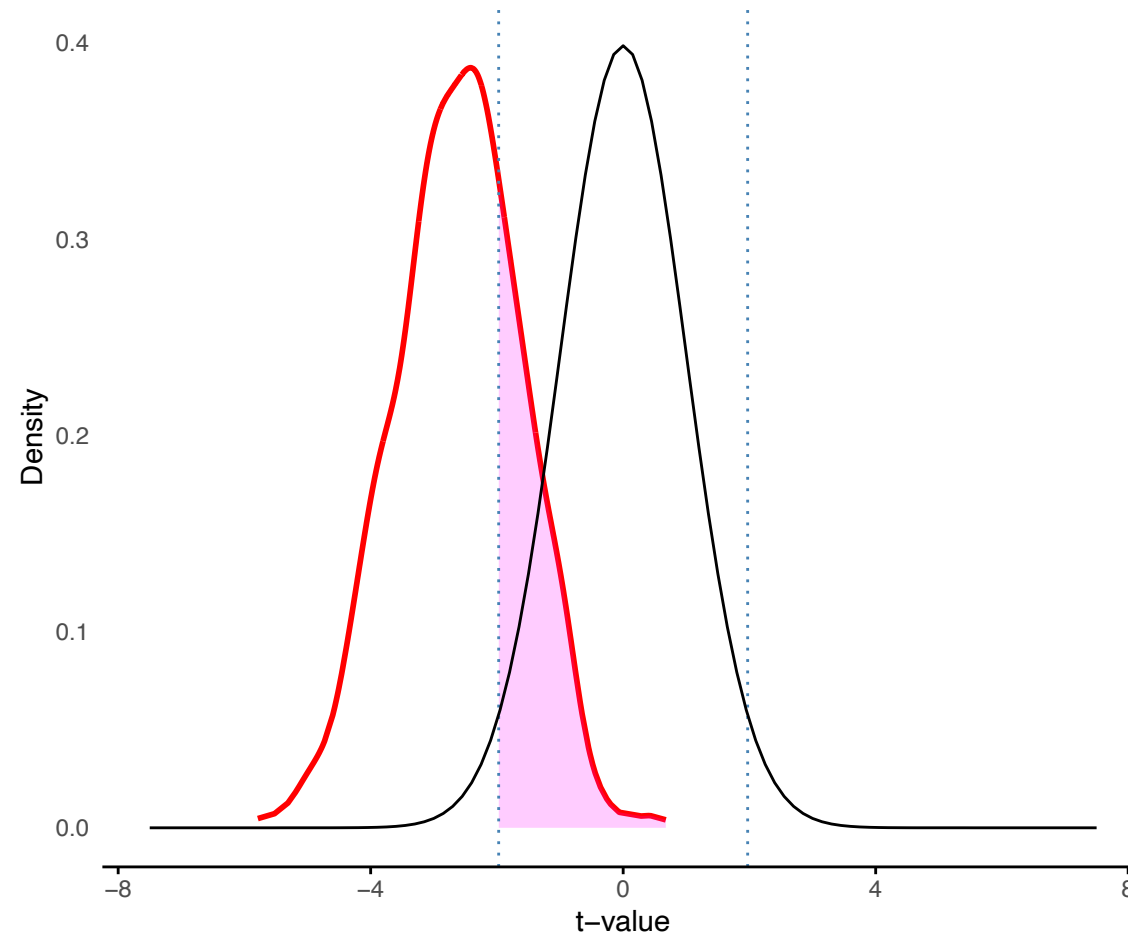
95% CI
[-4.12, -0.78]



Punktschätzer
 $\Delta = 2.5$



1000 mal wiederholt



Fehler beim Testen

schwanger!



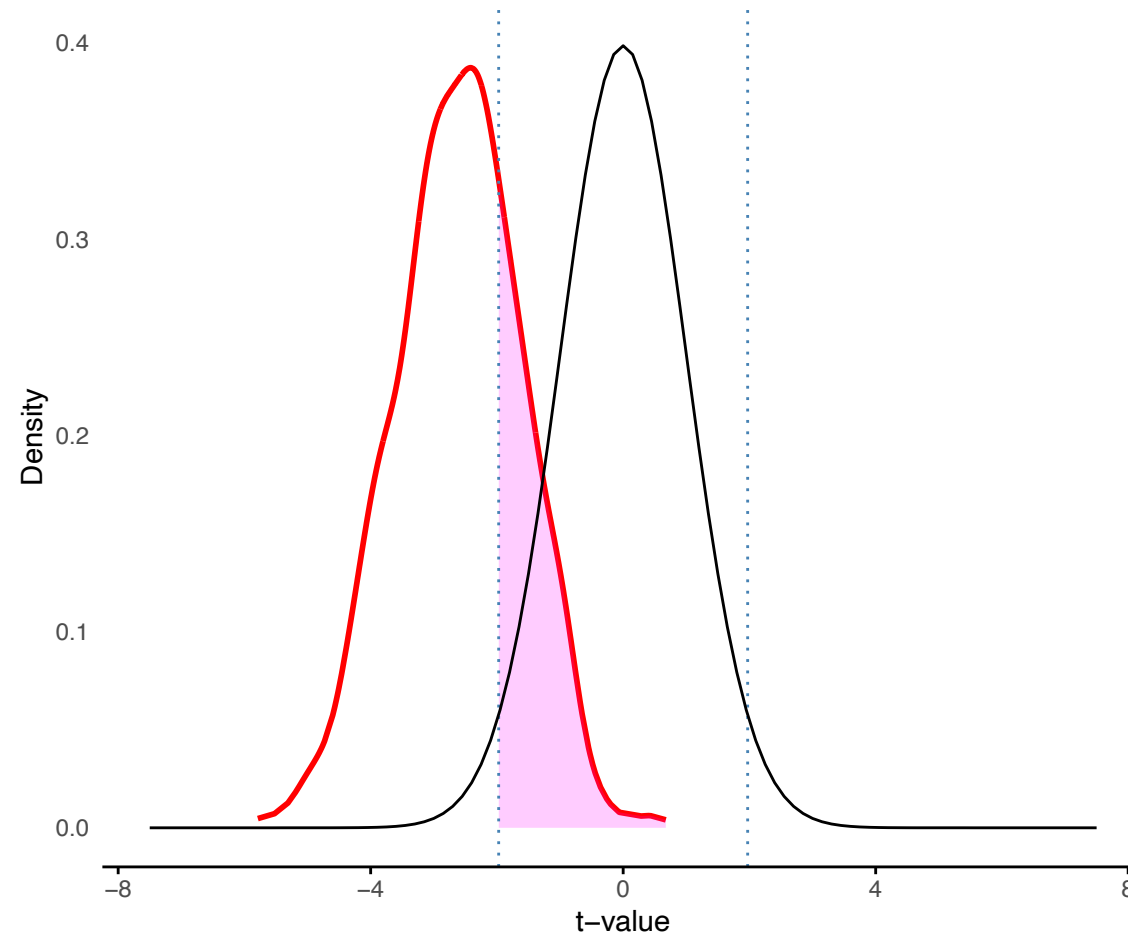
falsch positiv

Nicht
schwanger!

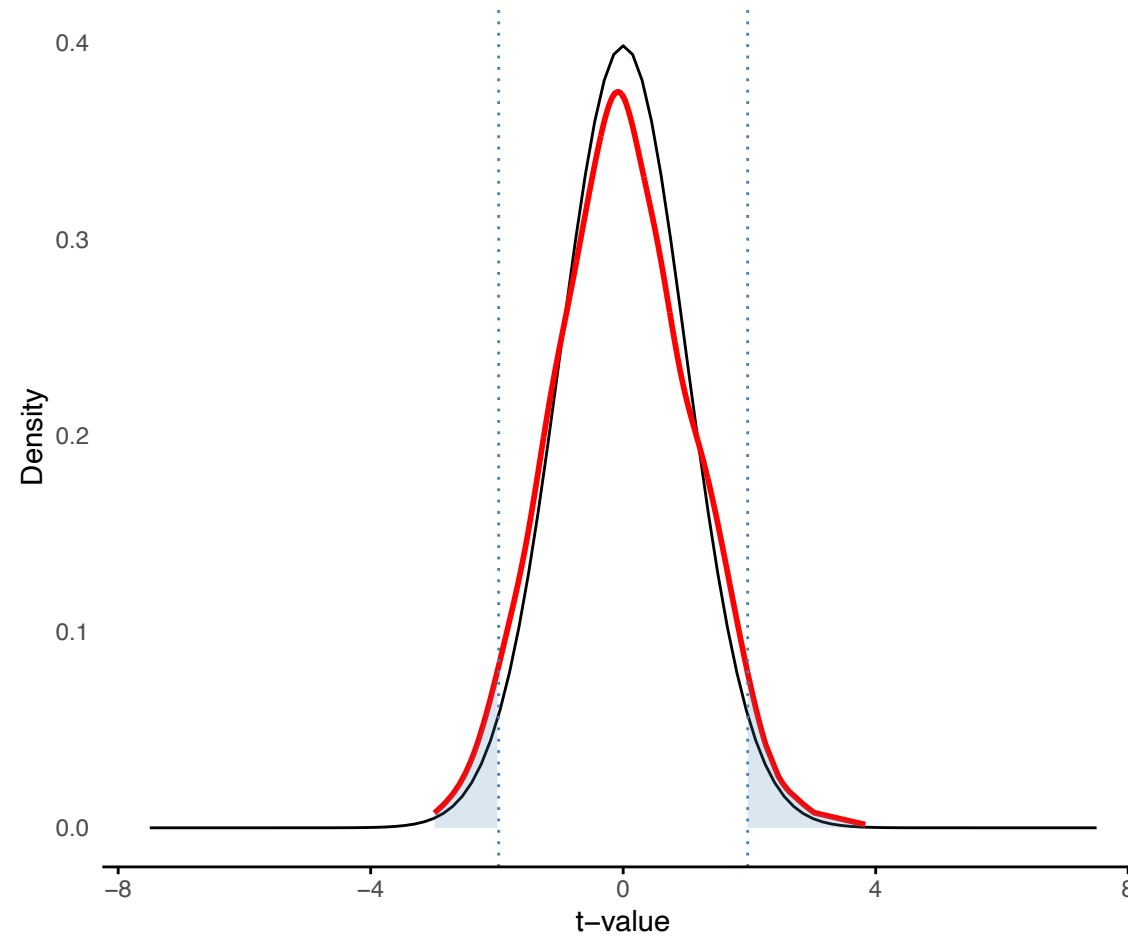


falsch negativ

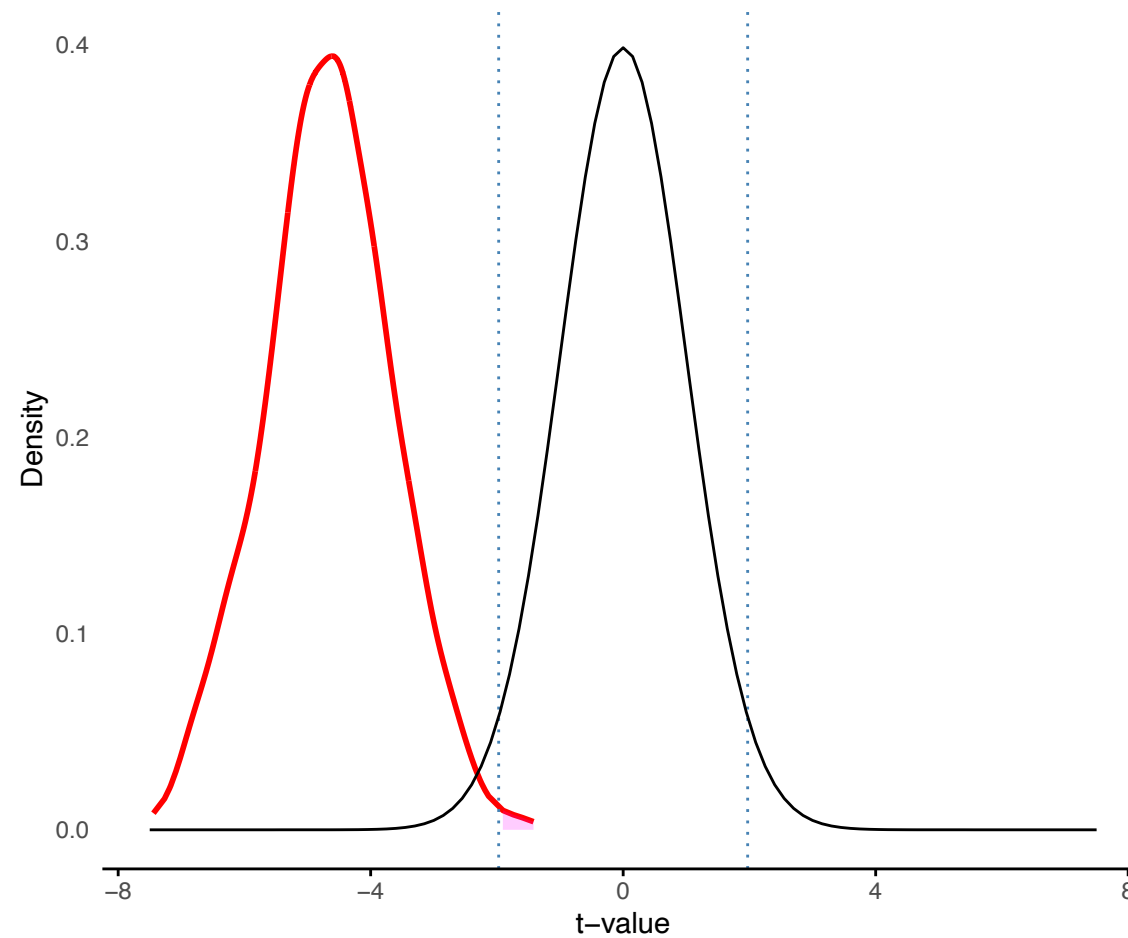
1000 mal wiederholt



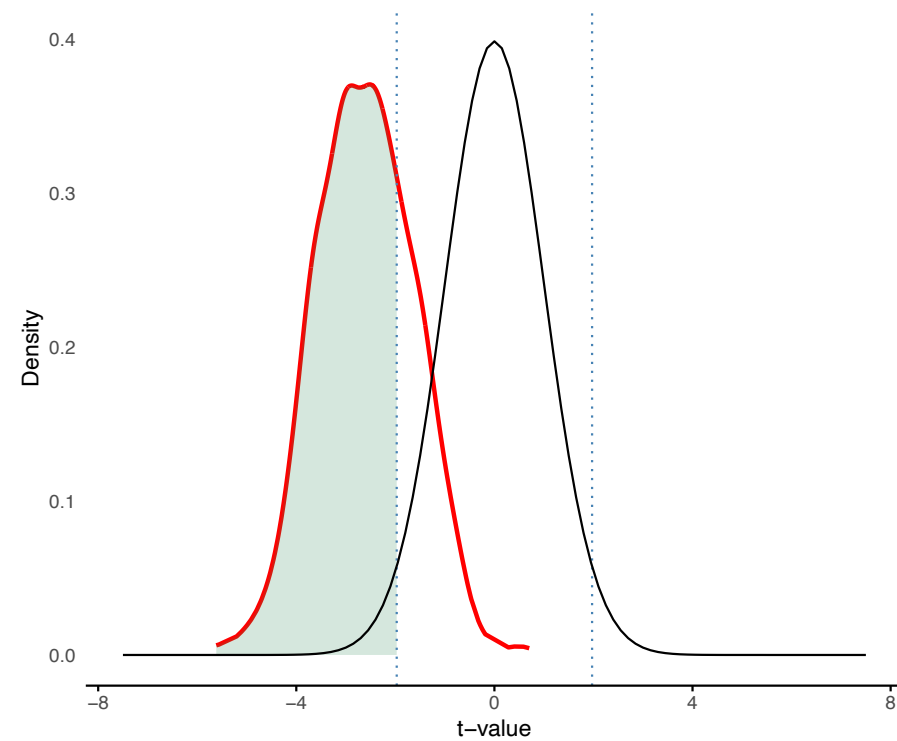
Diät ohne Effekt



Diät mit grossem Effekt



Power

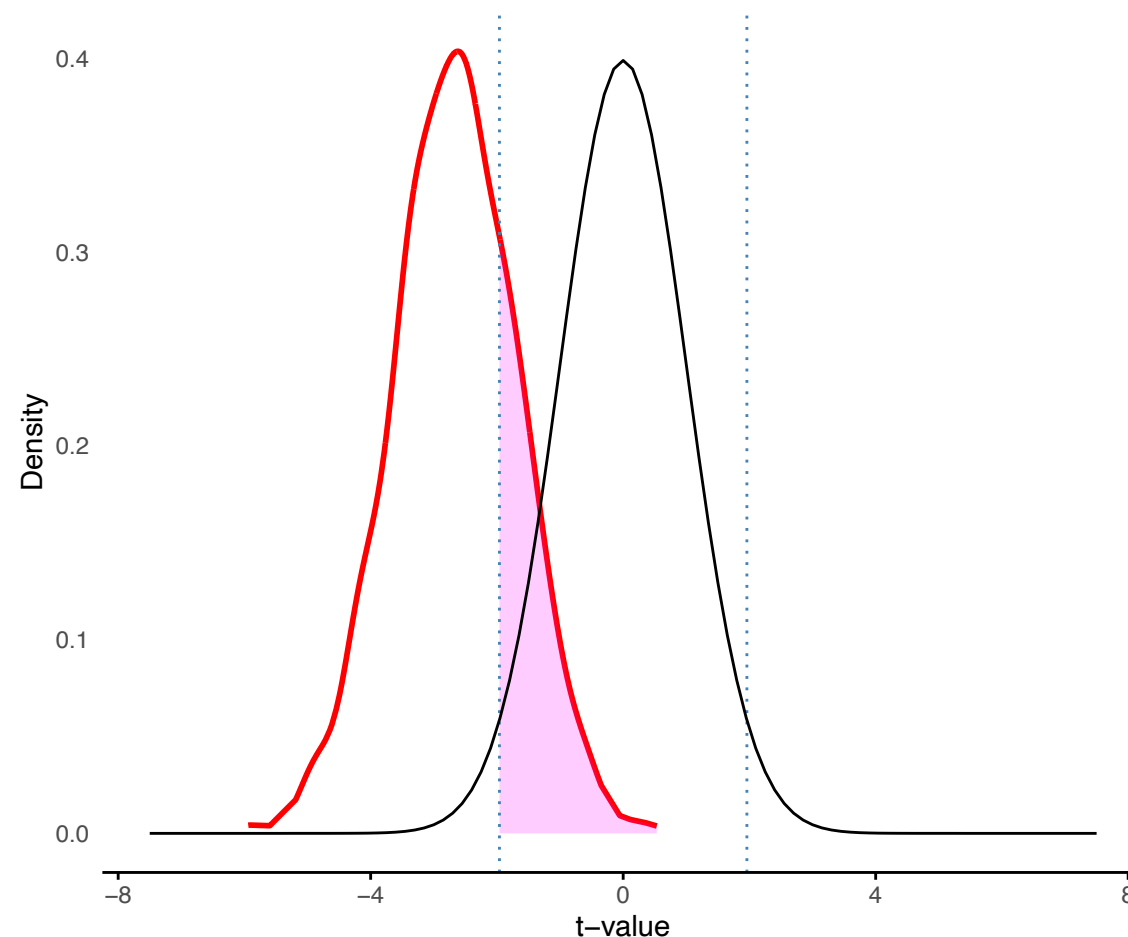


Die Sensitivität eines Studienstettings hängt ab von

- Effektgröße
- Stichprobengröße

Was zählt ist die Relevanz

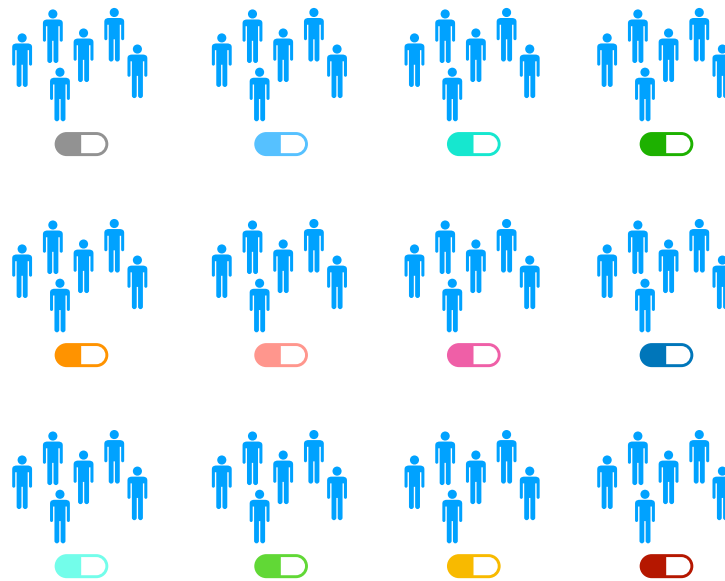
- Relevanter Effekt nicht signifikant (underpowered)
- Nicht relevanter Effekt aber signifikant



delta -0.25
N = 10000



Multiples Testen



- Beispiele: Sequentielles Testen, mehrere/viele Endpunkte, Mehrgruppenvergleiche.
- Erhöht die Gesamtwahrscheinlichkeit von falsch positiven Ergebnissen.
- Muss kontrolliert werden, reduziert Power.

Studiendesign

- Ziel: Maximierung der Sensitivität und Repräsentativität
- Definition der Endpunkte
- Maßnahmen:
 - Poweranalyse (N)
 - Planung repräsentativer, balancierter Stichproben (N, zufällig, Stratifizierung)
 - Ausschluss von Störgrößen und Voreingenommenheit (Stratifizierung, Blinding)
 - Abstimmung des statistischen Testverfahrens
- Vor-Registrierung zur Dokumentierung von Integrität.

RCT

- Goldstandard um den Einfluß einer Intervention auf eine bestimmte Meßgröße zu erfassen.
- Randomisierungsprozess, Stratifizierung = Imbalancevermeidung, Kontrollarm, Treatmentarm, Blinding.
- Cons: fehlende outcomes, noncompliance, dropouts, teuer, nicht geeignet für seltene Erkrankungen und akute, lebensgefährliche Erkrankungen

Beobachtende Studie

- Ökonomisch
- Zuteilung nicht zufällig und Kriterien, die diese Entscheidung bedingen, sind Störgrößen (systematische Fehler).
- Typen:
 - Prospective Kohortenstudie
 - Retrospective Kohortenstudie (treatment selection)
 - Fall-Kontroll-Studie (outcome selection)
- Cons: viel Voreingenommenheit, imbalance, nicht repräsentativ

Neurobiology of Disease

Cannabis Use Is Quantitatively Associated with Nucleus Accumbens and Amygdala Abnormalities in Young Adult Recreational Users

Morning Mix

Even casually smoking marijuana can change your brain, study says

Table 1. Participant demographics

	CON (<i>n</i> = 20)	MJ (<i>n</i> = 20)	<i>p</i> -value
Sex (M/F)	9 M/11 F	9 M/11 F	N/A
Age	20.7 (1.9)	21.3 (1.9)	0.30
Years of education	14.3 (3.4)	12.6 (4.8)	0.20
STAI ^a			
State	28.9 (7.94)	27.7 (7.38)	0.65
Trait	29.8 (7.32)	29.5 (5.56)	0.89
HAM-D ^b	0.80 (1.40) [range: 0–5]	1.10 (1.37) [range: 0–5]	0.50
TIP ^c			
Extroversion	10.9 (2.36)	10.7 (2.13)	0.78
Agreeableness	10.8 (2.47)	10.7 (1.81)	0.94
Conscientiousness	11.9 (2.08)	11.7 (2.13)	0.76
Emotional stability	10.5 (2.52)	11.4 (2.64)	0.27
Openness	12.1 (1.90)	12.4 (1.61)	0.57
Substance use			
Alcohol			
No. alcoholic drinks/week	2.64 (2.38)	5.09 (4.69)	0.10
AUDIT score	3.30 (1.78)	5.50 (2.21)	0.05
Cigarettes			
No. of occasional smokers ^d	0	7	N/A
No. of daily smokers	0	1	N/A
Marijuana			
No. days/week	0	3.83 (2.36)	N/A
No. joints/week	0	11.2 (9.61)	N/A
No. joints/occasion	0	1.80 (0.77)	N/A
No. smoking occasions/day	0	1.80 (0.70)	N/A
Age of onset (years)	—	16.6 (2.13)	N/A
Duration of use (years)	—	6.21 (3.43)	N/A

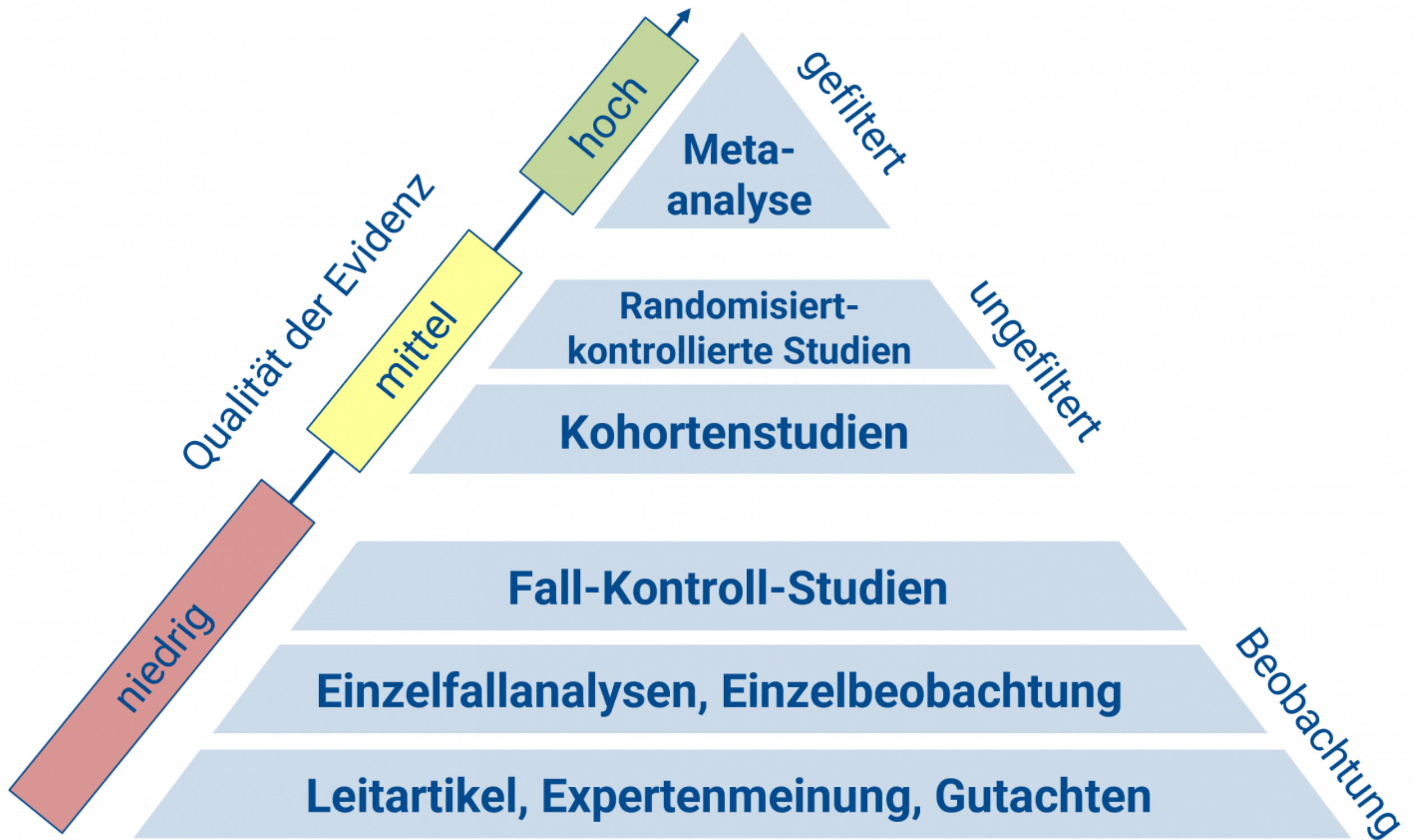
All values are expressed in means and SDs. CON, controls; MJ, marijuana users.

^aState Trait Anxiety Inventory Form (Spielberger et al., 1983).

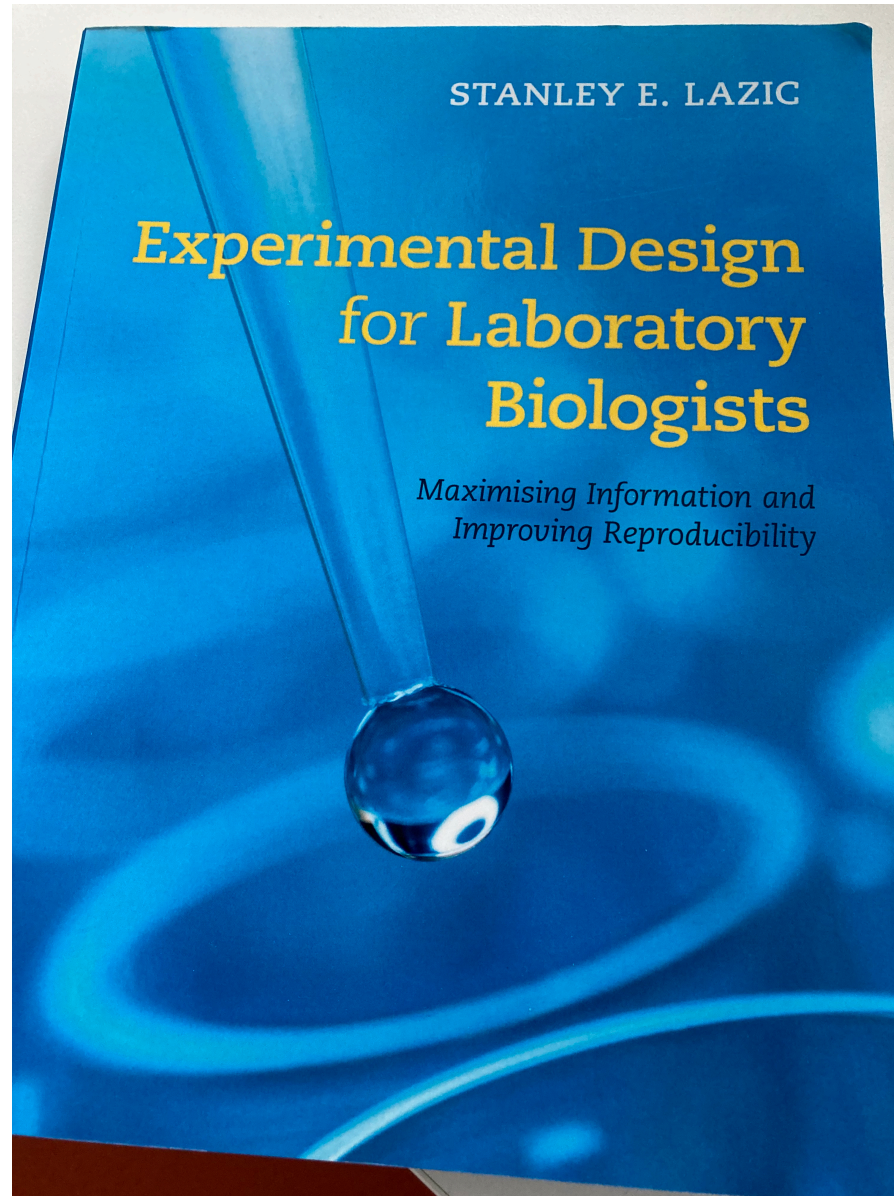
^bHamilton Depression Rating Scale (Hamilton, 1960).

^cTen-Item Personality Inventory (Gosling et al., 2003).

^dOccasional smokers reported from 1 cigarette/week to 1 cigarette every 3 months.



Im Labor



Flowchart experimentelle Studie

- Formuliere *Hypothese* vor der Messung
- Entwerfe ein Experiment zum testen der Hypothese
 - Vergleichend (idealerweise 2 Bedingungen, Zweigruppenvergleich)
 - Definiere Endpunkt und dessen Beziehung zur Hypothese.
 - Bestimme statistisches Testverfahren und Stichprobengröße
 - Identifiziere potentielle Fehlerquellen und Kontrollprozeduren
- Führe das Experiment vollständig durch
- Analysiere die Daten
- Validiere das Ergebnis mit unabhängiger/orthogonaler Studie

Checkliste Studien Lesen

- Stichprobe repräsentativ/bedeutsam?
- Wahl des Endpunktes bedeutsam?
- Singuläre Studien mit Vorsicht genießen, falsch positiv?
- Kleines N, kleine Power, falsch negativ?
- Signifikant nicht notwendigerweise relevant.
- P alleine uninformativ. Konfidenzintervall?
- Vor-registriert?

Statistik lesen

A Randomized Trial of Hydroxychloroquine as Postexposure Prophylaxis for Covid-19

D.R. Boulware, M.F. Pullen, A.S. Bangdiwala, K.A. Pastick, S.M. Lofgren, E.C. Okafor, C.P. Skipper, A.A. Nascene, M.R. Nicol, M. Abassi, N.W. Engen, M.P. Cheng, D. LaBar, S.A. Lother, L.J. MacKenzie, G. Drobot, N. Marten, R. Zarychanski, L.E. Kelly, I.S. Schwartz, E.G. McDonald, R. Rajasingham, T.C. Lee, and K.H. Hulsiek

ABSTRACT

BACKGROUND

Coronavirus disease 2019 (Covid-19) occurs after exposure to severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). For persons who are exposed, the standard of care is observation and quarantine. Whether hydroxychloroquine can prevent symptomatic infection after SARS-CoV-2 exposure is unknown.

METHODS

We conducted a randomized, double-blind, placebo-controlled trial across the United States and parts of Canada testing hydroxychloroquine as postexposure prophylaxis. We enrolled adults who had household or occupational exposure to someone with confirmed Covid-19 at a distance of less than 6 ft for more than 10 minutes while wearing neither a face mask nor an eye shield (high-risk exposure) or while wearing a face mask but no eye shield (moderate-risk exposure). Within 4 days after exposure, we randomly assigned participants to receive either placebo or hydroxychloroquine (800 mg once, followed by 600 mg in 6 to 8 hours, then 600 mg daily for 4 additional days). The primary outcome was the incidence of either laboratory-confirmed Covid-19 or illness compatible with Covid-19 within 14 days.

RESULTS

We enrolled 821 asymptomatic participants. Overall, 87.6% of the participants (719 of 821) reported a high-risk exposure to a confirmed Covid-19 contact. The incidence of new illness compatible with Covid-19 did not differ significantly between participants receiving hydroxychloroquine (49 of 414 [11.8%]) and those receiving placebo (58 of 407 [14.3%]); the absolute difference was -2.4 percentage points (95% confidence interval, -7.0 to 2.2 ; $P=0.35$). Side effects were more common with hydroxychloroquine than with placebo (40.1% vs. 16.8%), but no serious adverse reactions were reported.

CONCLUSIONS

After high-risk or moderate-risk exposure to Covid-19, hydroxychloroquine did not prevent illness compatible with Covid-19 or confirmed infection when used as postexposure prophylaxis within 4 days after exposure. (Funded by David Baszucki and Jan Ellison Baszucki and others; ClinicalTrials.gov number, NCT04308668.)

The authors' full names, academic degrees, and affiliations are listed in the Appendix. Address reprint requests to Dr. Boulware at the University of Minnesota, 689 23rd Ave., Minneapolis, MN 55455, or at boulw001@umn.edu.

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Statistik lesen

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Hypothese

Design

N

Endpunkt

Planung

Vorregistrierung

N

Stichprobe

Durchführung

Statistische Analyse

Interpretation

Ergebnis